IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE/EIE PART III/IV: MEng, BEng and ACGI

DIGITAL SIGNAL PROCESSING

Thursday, 12 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Stathaki
Second Marker(s) : W. Dai

© Imperial College London

1. (a) Consider the causal transfer function:

$$H_M(z) = \frac{B(z)}{A(z)} = \frac{\sum_{n=0}^{n=M} b[n]z^{-n}}{\sum_{n=0}^{n=M} a[n]z^{-n}}, M > 0$$

where $b[n] = a^*[M-n]$. The above transfer function is an allpass filter.

(i) Show that the phase of $H_M(e^{j\omega})$ is $\angle H_M(e^{j\omega}) = -\omega M - 2\angle A(e^{j\omega})$, where $\angle A(e^{j\omega})$ is the phase of the polynomial $A(e^{j\omega})$. **[3]**

(ii) Consider a causal, real, stable, allpass transfer function $H(z)$ given as:

$$H(z) = \frac{-pz^{-q} + z^{-1-q}}{1 - pz^{-1}}$$

Sketch the pole-zero plot of $H(z)$ if $p = 0.5$, where q is a positive integer. **[3]**

(iii) Determine the phase response of $H(z)$ as a function of the pole p and the parameter q using the result from part (a)(i) above. **[3]**

(b) Consider a second-order all-pass filter with a transfer function:

$$H(z) = \frac{a_2 + a_1 z^{-1} + a_0 z^{-2}}{a_0 + a_1 z^{-1} + a_2 z^{-2}}$$

Suppose that $a_0 = 1,\ a_1 = -3a/4$ and $a_2 = a^2/8$ where a is a real parameter.

(i) Determine the condition on a for the filter to be stable. **[2]**

(ii) Find the phase response of $H(z)$ as a function of the poles of $H(z)$. **[3]**

(iii) Calculate the phase introduced by this filter at $\omega = 0$ and $\omega = \pi$. **[2]**

(c) The average group delay of an IIR stable system is $n - m$ where n is the number of poles and m is the number of zeros of the system located inside the unit circle.

(i) Demonstrate that the factors $(z - q)$ and $(q^*z - 1)$ have the same amplitude response. Use this result to show that inverting a zero originally located inside the unit circle to a position outside the unit circle increases the group delay by 1. Here, q is a complex number and q^* represents its complex conjugate. **[3]**

(ii) Using the conclusion from part c(i) above, identify which filter, among all filters with the same amplitude response, achieves the minimum group delay. Such a filter is referred to as a minimum-phase filter. **[3]**

(iii) Explain why linear-phase filters cannot be categorized as minimum-phase filters. **[3]**

2. (a) (i) Demonstrate that antisymmetric linear-phase Finite Impulse Response (FIR) filters of order N with a transfer function $H(z)$ (**Type 4**), satisfy the Antimirror Image Polynomial relationship (AIP) given below:

$$H(z) = -z^{-(N-1)}H(z^{-1})$$

[4]

(ii) Investigate the types of filters (e.g., low-pass, high-pass) that can be implemented using **Type 4** linear-phase Finite Impulse Response (FIR) transfer functions. **[4]**

(iii) Prove that a Type 4 linear-phase Finite Impulse Response (FIR) transfer function, $H(z)$, can be represented in the following form:

$$H(z) = \left(\sum_{n=0}^{\frac{N}{2}-1} h[n](z^{-n} - z^{n-N+1}) \right)$$

[4]

(b) Let $H(z)$ represent the transfer function of a real-coefficient, linear-phase FIR filter. The filter has zeros located at specific points in the z −plane, with a subset of these zeros given as:

$$\left\{1, 0.5e^{j\frac{\pi}{6}}, -\frac{1}{5}, j\right\}$$

(i) Since the zeros of a linear-phase FIR filter must exist in particular configurations, we deduce that the filter $H(z)$ must include additional zeros beyond those provided. Determine the additional zeros of $H(z)$ for each possible type of linear-phase FIR filter with the minimum allowable length. **[6]**

(ii) What is the filter type of the minimum length? **[3]**

(iii) Sketch the zero diagram of the filter identified in (ii) above on the z−plane. **[4]**

3. (a) The bilinear transformation from the s −plane to the z −plane is given by

$$z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s}$$

Here, T is the sampling period of the signal.

(i) Illustrate, using mathematical relationships, how the bilinear transformation and its inverse map each point $s = \sigma + j\Omega$ in the s −plane to a corresponding point $z = re^{j\omega}$ in the z −plane, and vice versa. **[3]**

(ii) Prove that the relationship between continuous-time angular frequency Ω and discrete-time angular frequency ω under the bilinear transformation is generally non-linear. Additionally, demonstrate that this mapping becomes approximately linear at low frequencies.
[**Hint:** Use the small-angle approximation for the tangent function, $\tan(x) \approx x$ when $x \to 0$]. **[3]**

(b) The transfer function of an ideal analogue integrator is given by:

$$H_a(s) = \frac{1}{s}$$

A digital integrator can be derived from an analogue integrator by applying the bilinear transformation.

(i) Determine the amplitude and phase response of the analogue integrator and provide a rough sketch to illustrate the results. **[3]**

(ii) Derive the input-output relationship for the digital integrator in the time domain. For simplicity use $T = 2$ for the sampling period. **[3]**

(iii) Estimate the frequency response of the digital integrator. **[3]**

(iv) Create a rough sketch of the amplitude and phase responses of the digital integrator. **[3]**

(v) Compare the amplitude and phase responses of the analogue and digital integrators, with a focus on demonstrating that the amplitude responses are more closely aligned at low frequencies compared to high frequencies. **[3]**

(vi) Demonstrate that the pole of the digital integrator, located on the unit circle, causes its response to become unbounded for inputs with a DC component. Use the input signal $x[n] = u[n]$, where $u[n]$ is the unit step function, to illustrate this behaviour. **[4]**

4. (a) Examine the multirate configuration illustrated in **Figure 4.1** below. In this setup, $H_0(z), H_1(z)$, $H_2(z)$ and $H_3(z)$ are filters, each possessing ideal, zero-phase characteristics and real coefficients. Their corresponding frequency responses are outlined as follows:

$$H_0(e^{j\omega}) = u(\omega) - u(\omega - \frac{\pi}{6})$$

$$H_1(e^{j\omega}) = u(\omega - \frac{\pi}{6}) - u(\omega - \frac{\pi}{2})$$

$$H_2(e^{j\omega}) = u\left(\omega - \frac{\pi}{2}\right) - u\left(\omega - \frac{5\pi}{6}\right)$$

$$H_3(e^{j\omega}) = u\left(\omega - \frac{5\pi}{6}\right) - u(\omega - \pi)$$

The function $u(\omega)$ is the unit step function, defined as

$$u(\omega) = \begin{cases} 1 & \omega \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

If the input is a real sequence with a Discrete Time Fourier Transform

$$X(e^{j\omega}) = (\frac{\pi}{6} - \frac{6\omega^2}{\pi})\left(u(\omega) - u\left(\omega - \frac{\pi}{6}\right)\right)$$

sketch the Discrete Time Fourier Transform of the outputs $y_0[n], y_1[n], y_2[n]$ and $y_3[n]$. It's important to note that all frequency domain representations mentioned are limited to the range $[0, \pi]$. **[15]**

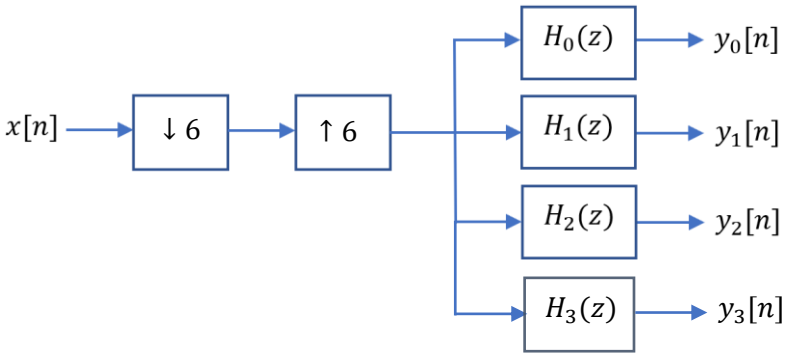


Figure 4.1

(b) Analyse the structure of **Figure 4.2** below and determine its input-output relations. **[10]**

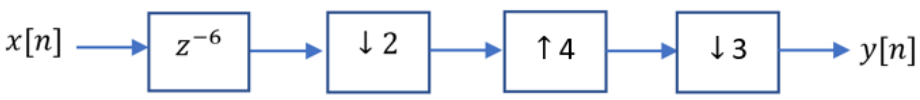


Figure 4.2